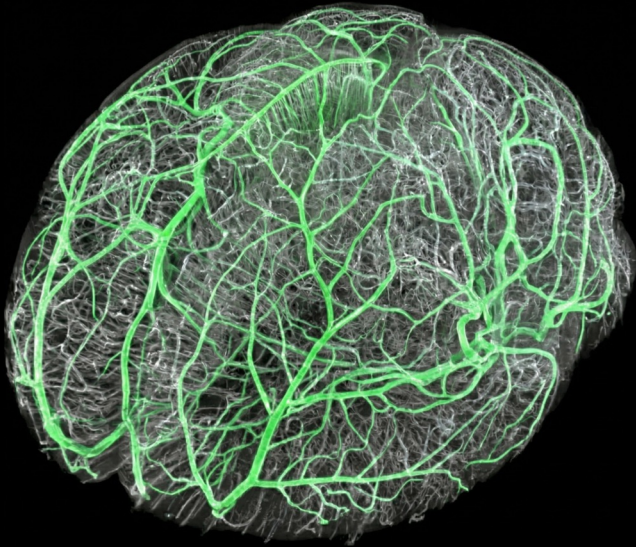
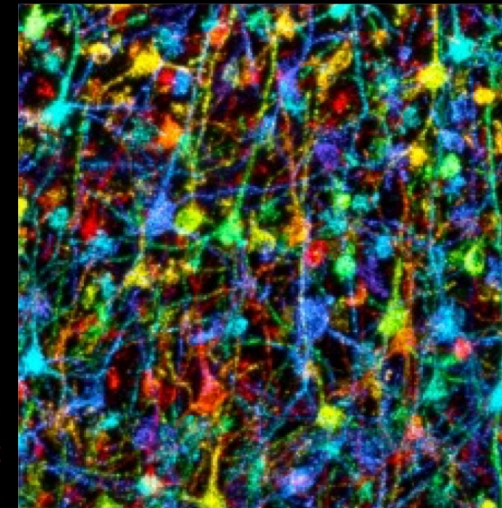


Opportunities:

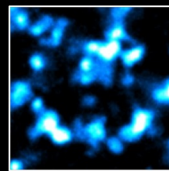
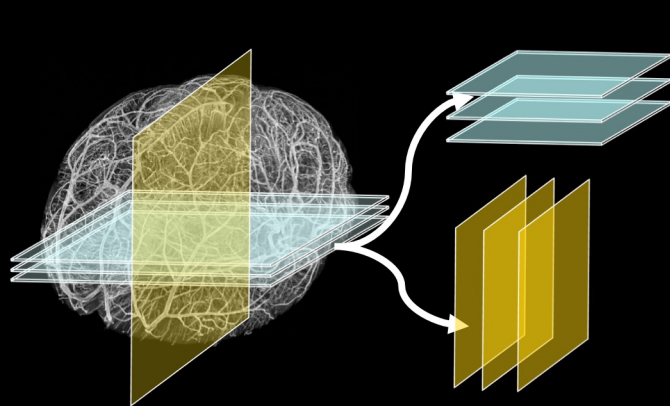
Expansion microscopy can reveal nanoscale neural structures and disease markers in whole organ level — hidden from conventional pathology (e.g. neuron–protein interactions, circuit mapping, neurodegeneration biomarkers).



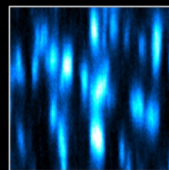
Difficulty in Scalability A single brain specimen generates 100+ TB of raw data for meaningful neuronal tracing - too large to store, share across sites, or analyze at the population level. Critically, existing super-resolution methods make storage worse, not better.



A fully resolved Cornu of Ammonis 3 (CA3) at 30nm ~2 PB data!



High Resolution Lateral



Low Resolution Axial

Difficulty in Interpretability Organ level acquisition requires low-NA, wide-field optics for speed, which degrades axial resolution.

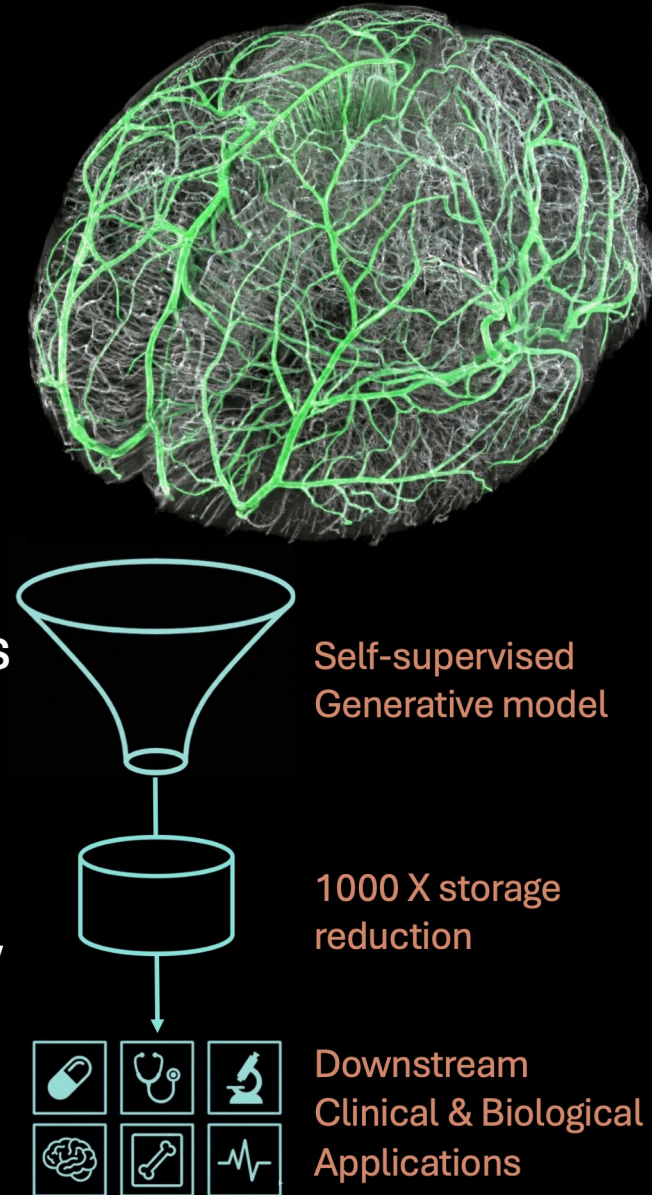
Biomarkers appear fused in the axial view, neuron–protein overlaps become unresolvable, and quantitative 3D morphometry is biased.

What we did

A self-supervised AI model that both resolves the blurry axis (up to 8×) and compresses data (~1000× storage reduction) — validated on human surgical brain tissue.

Self-supervised no labeled ground truth is needed —learns directly from the raw acquisition.

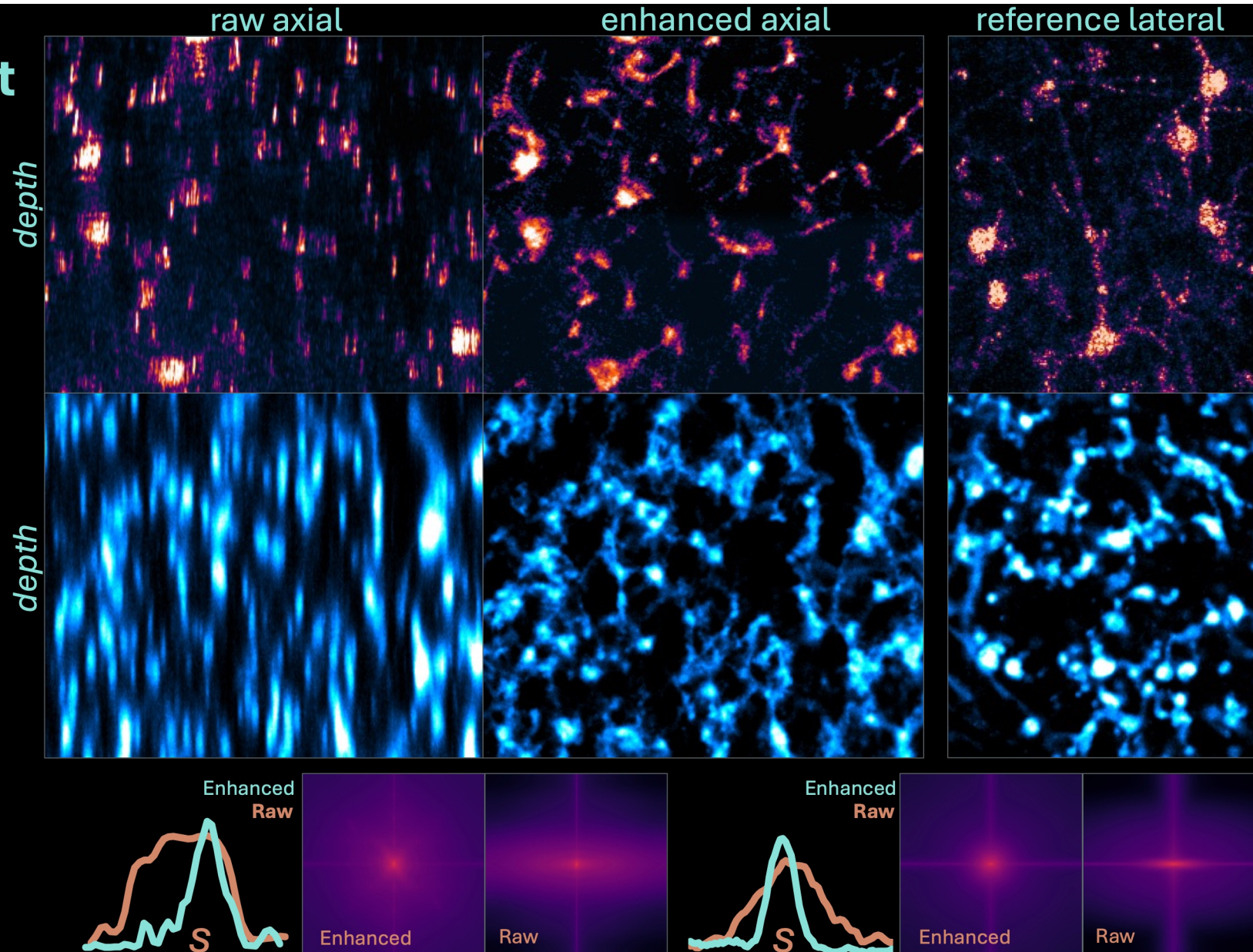
High compression ratio means a whole-brain dataset that would cost petabytes to store and days to transfer can now fit on a single drive and be reconstructed on demand.



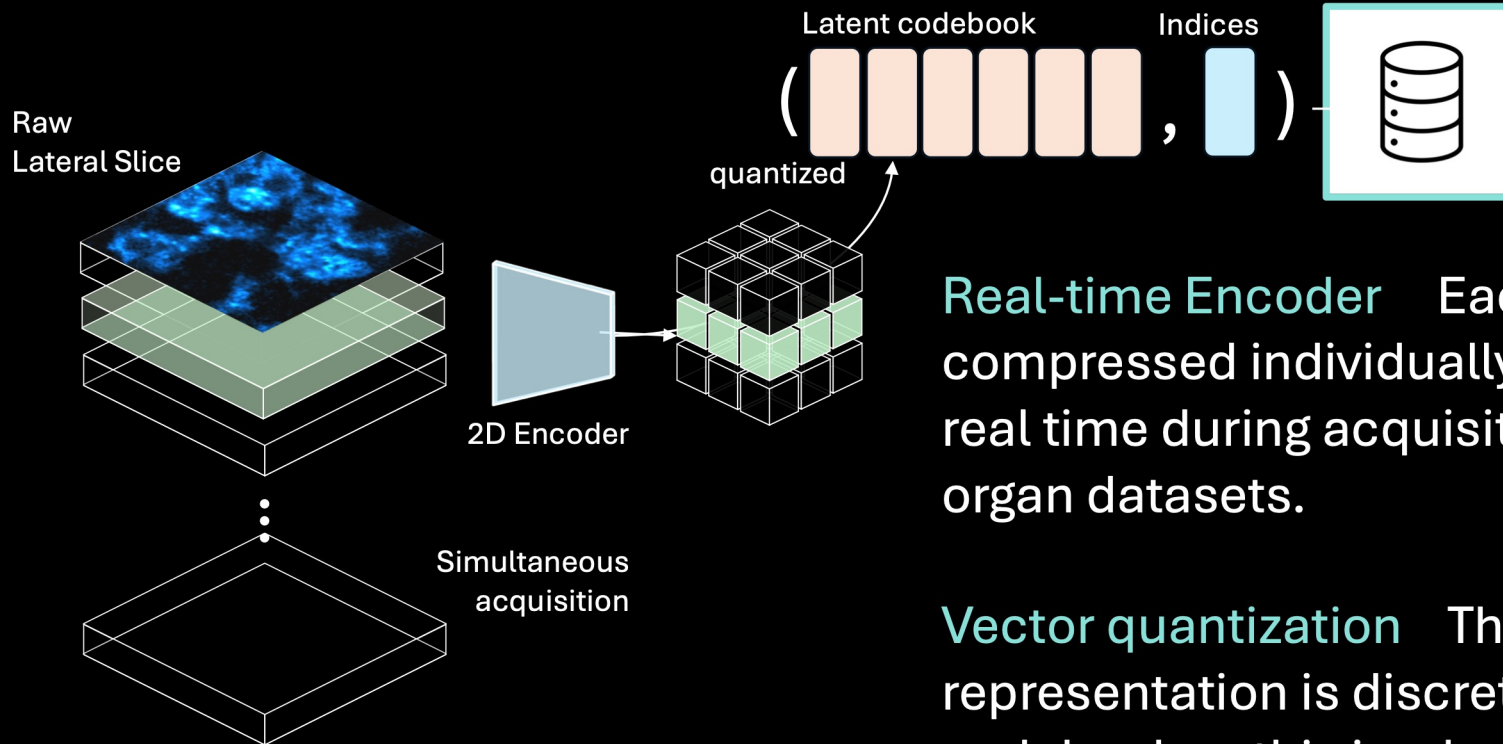
AI-Enhancement 3D Imaging biomarker

AI-enhanced axial
views recover neural
structures
indistinguishable
from high-resolution
lateral references.

Turning previously
unreadable depth
views into
quantifiable 3D data.



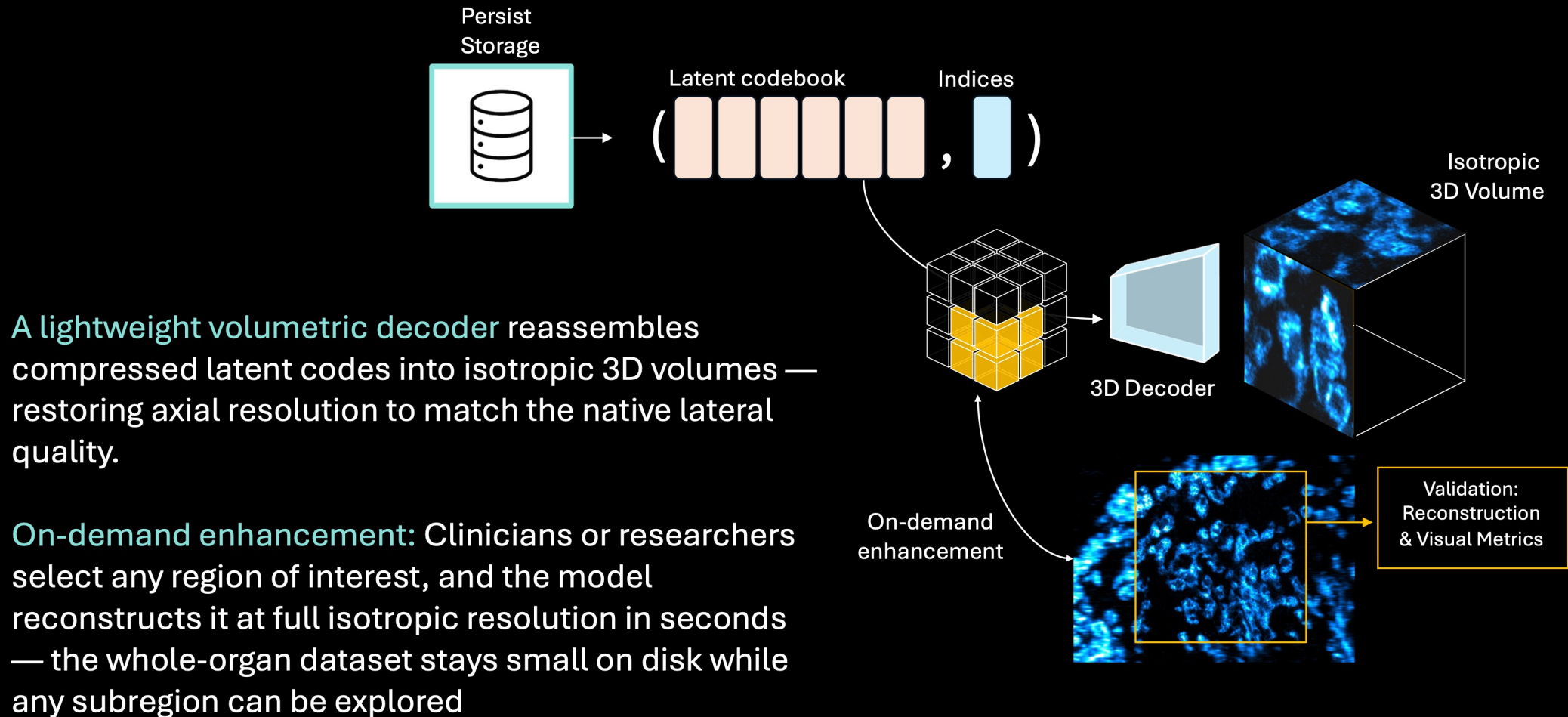
Foundation Model of High-throughput 3D Tissue Imaging



Real-time Encoder Each raw lateral slice is compressed individually - so encoding runs in real time during acquisition, even for whole-organ datasets.

Vector quantization The compressed representation is discretized into a learned codebook — this is what achieves the 128× compression while preserving the biological detail needed for downstream reconstruction.

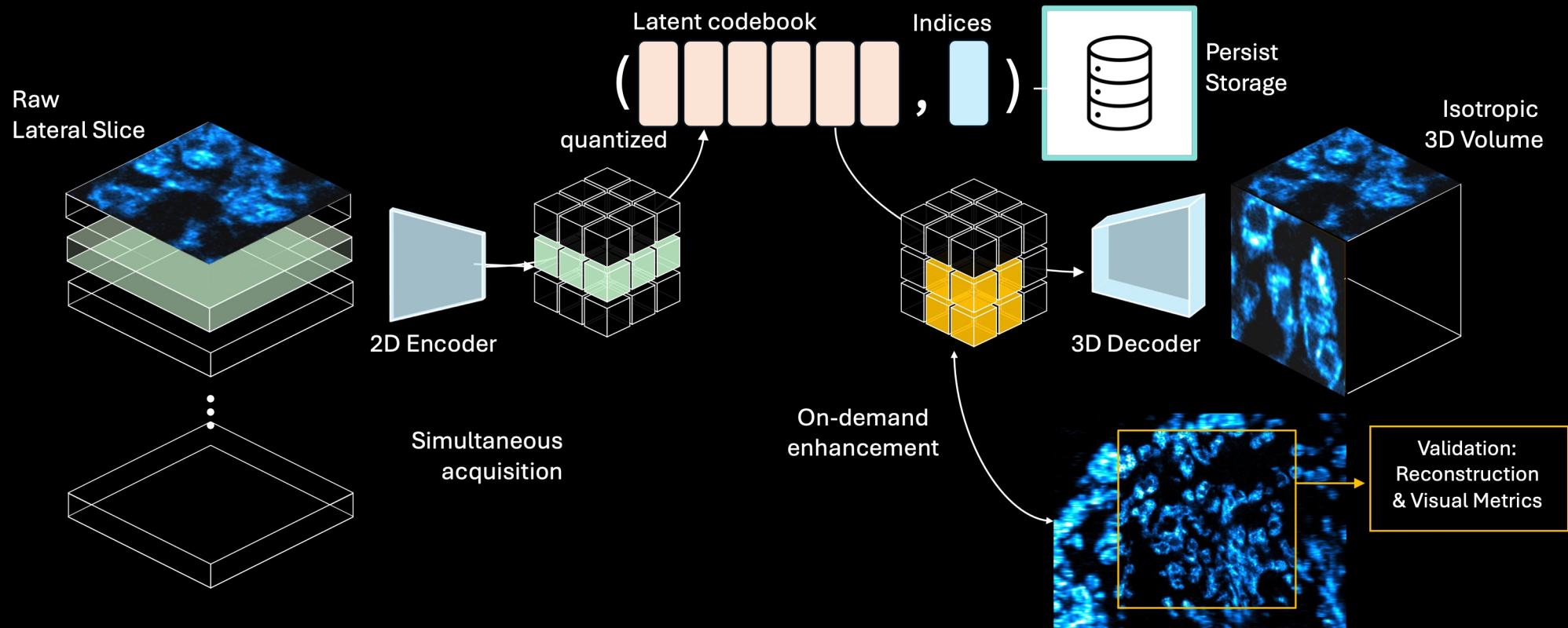
Foundation Model of High-throughput 3D Tissue Imaging



Foundation Model of High-throughput 3D Tissue Imaging

Data Compression Module:
up to 128 X compression ratio

Image Super-resolution Module
up to 8 X resolution enhancement

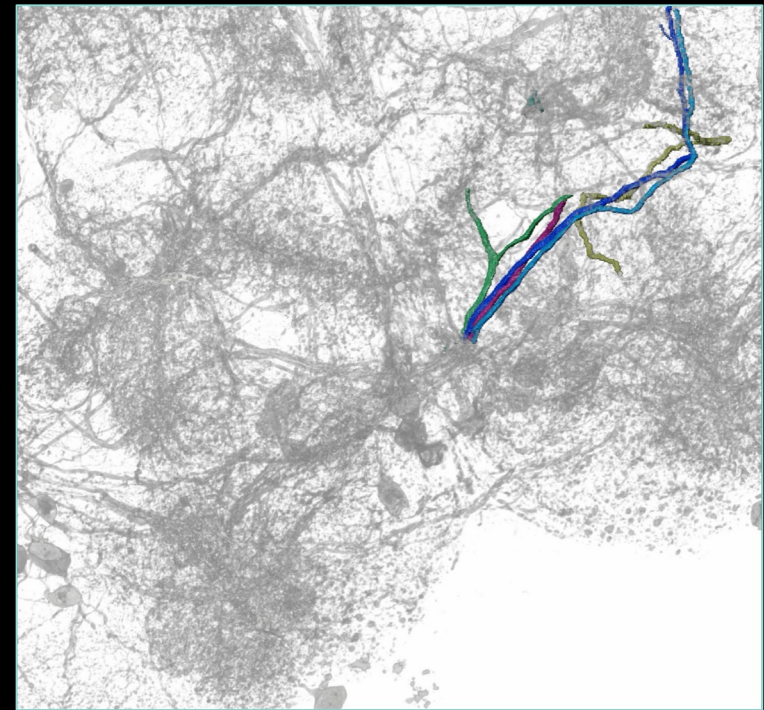
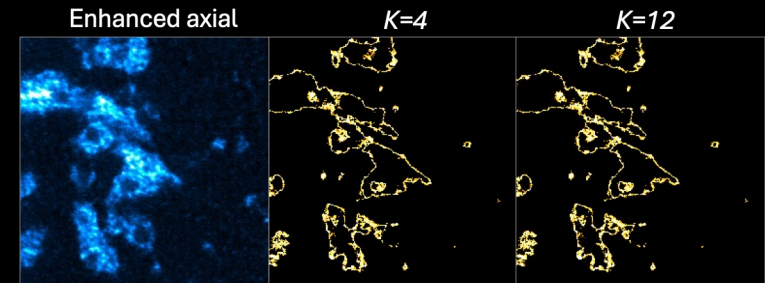
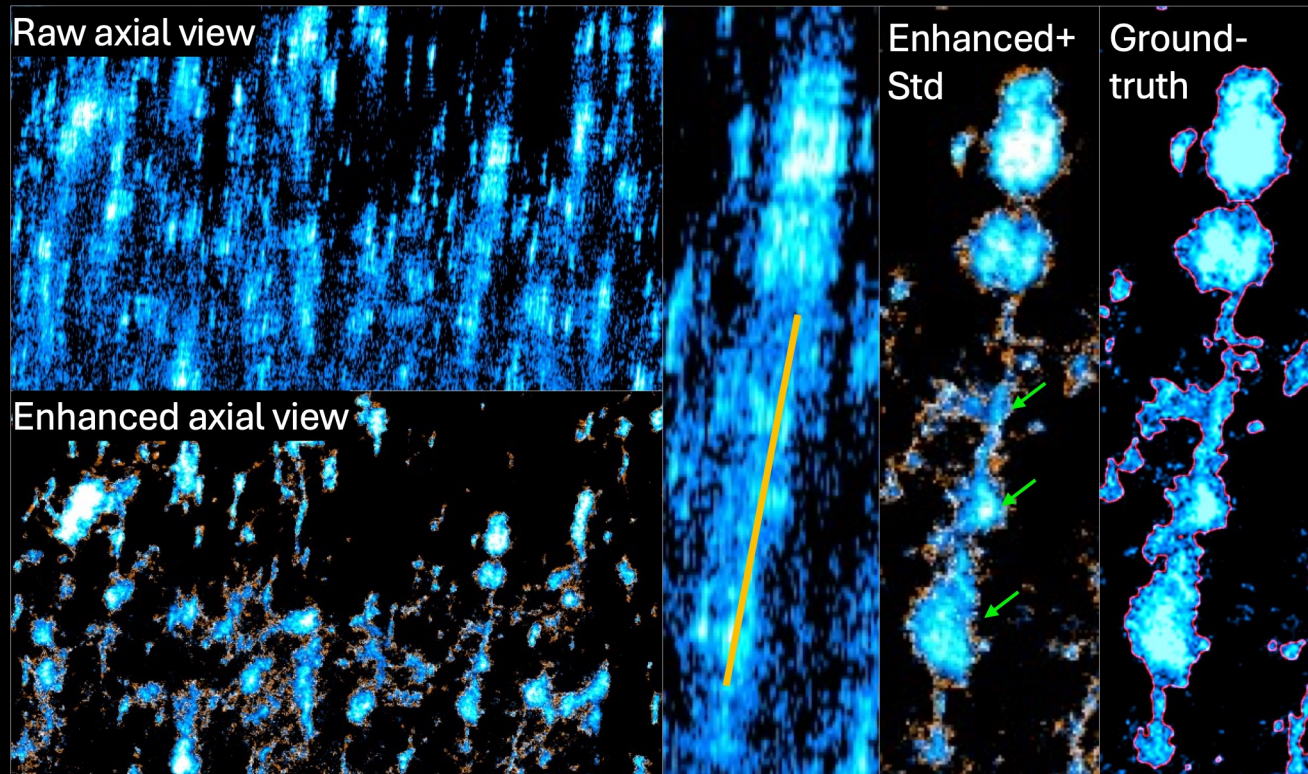


Deliverables: up to **~1000 X** efficiency in high-resolution imaging biomarkers identification (PB > TB scale)

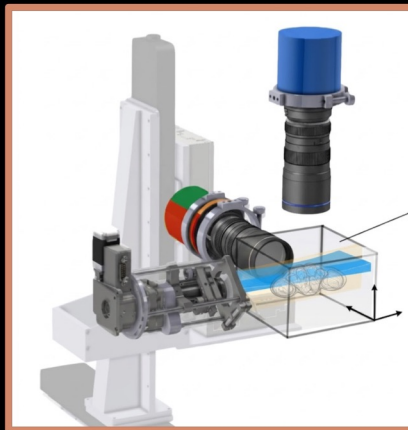
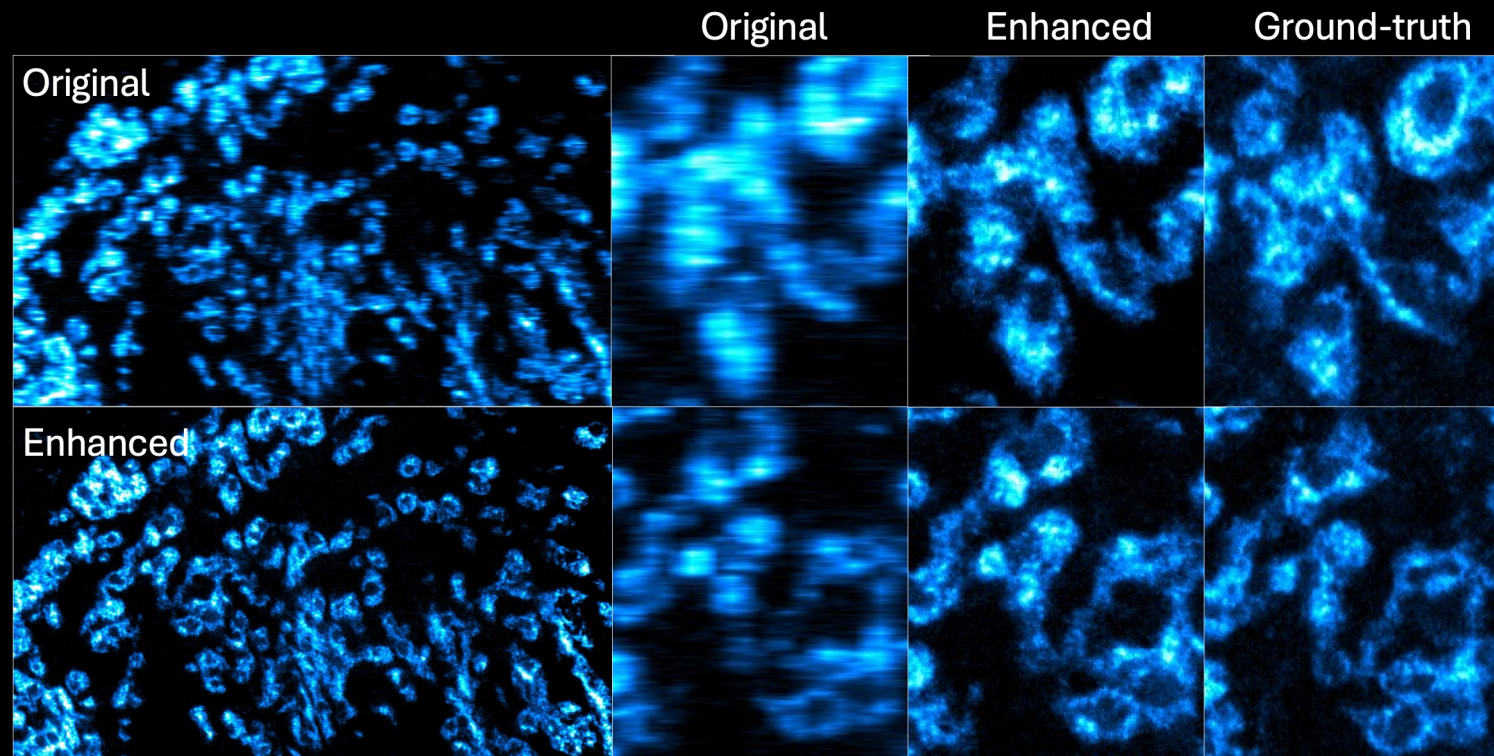
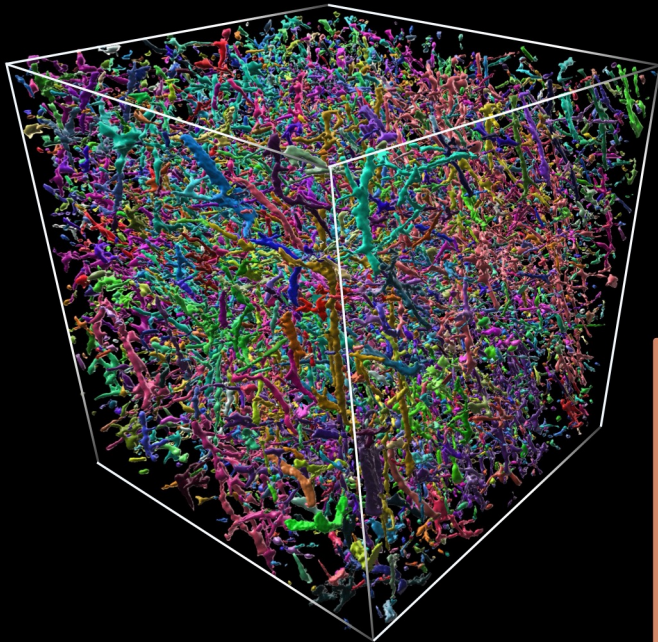
Self-supervised segmentation for neuronal tracing

Monte-Carlo inference enabled
unsupervised segmentation
No annotation needed

AI-enabled large scale neuronal tracing



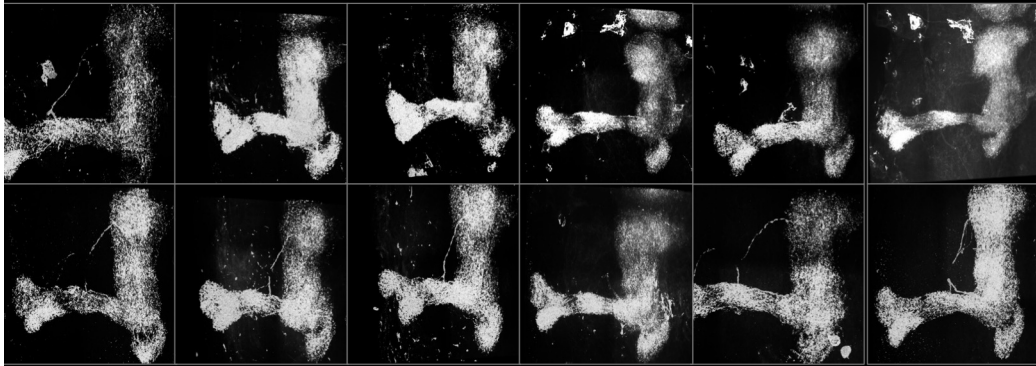
Translation to human brain cell study



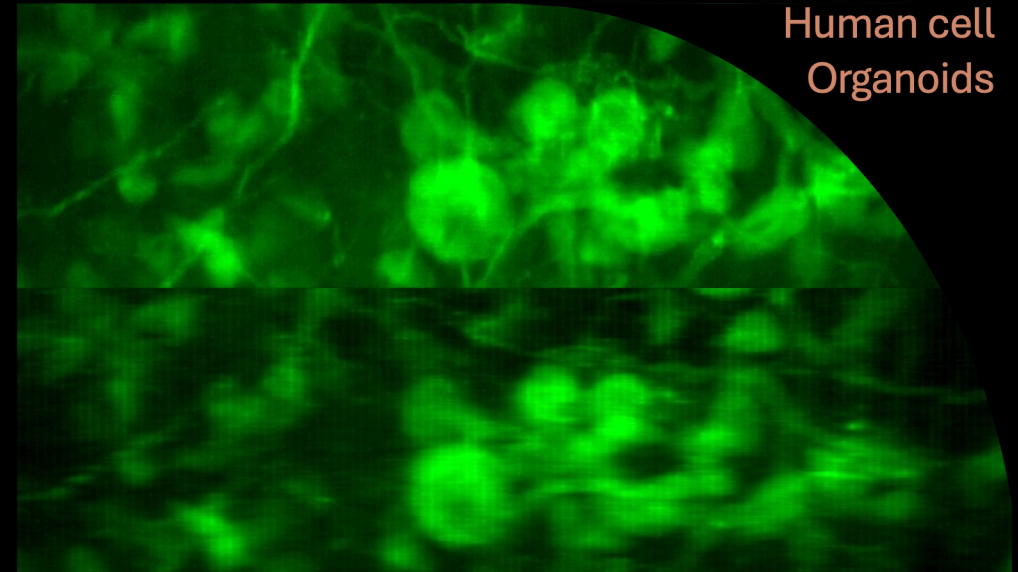
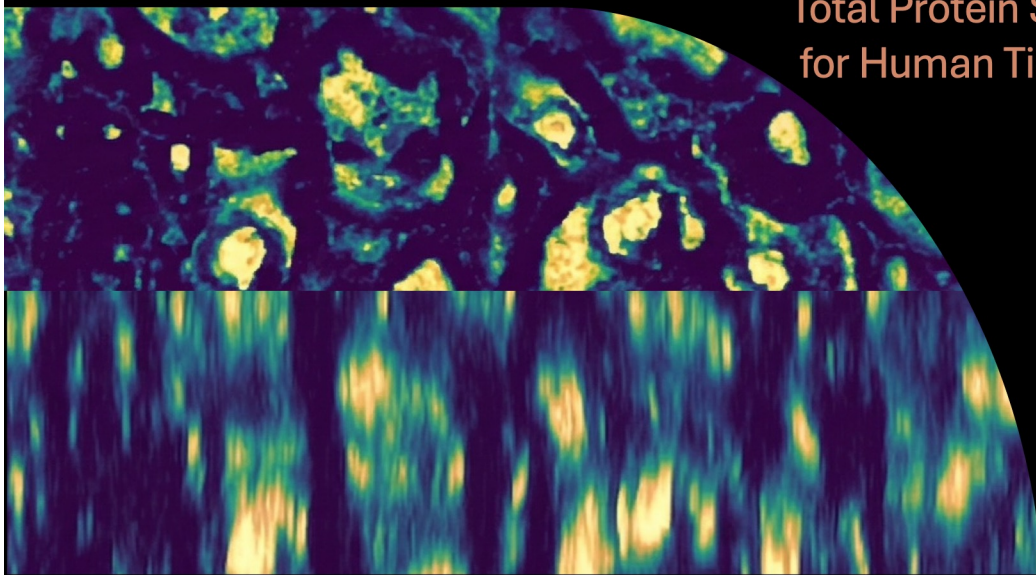
Golgi stain revealing complete individual neuron morphology—in intact human tissue. Essential for mapping neural structures linked to Alzheimer's, schizophrenia, and autism.

Our model makes large scale Golgi datasets storable, shareable, and reconstructible across institutions — the infrastructure required for human brain studies.

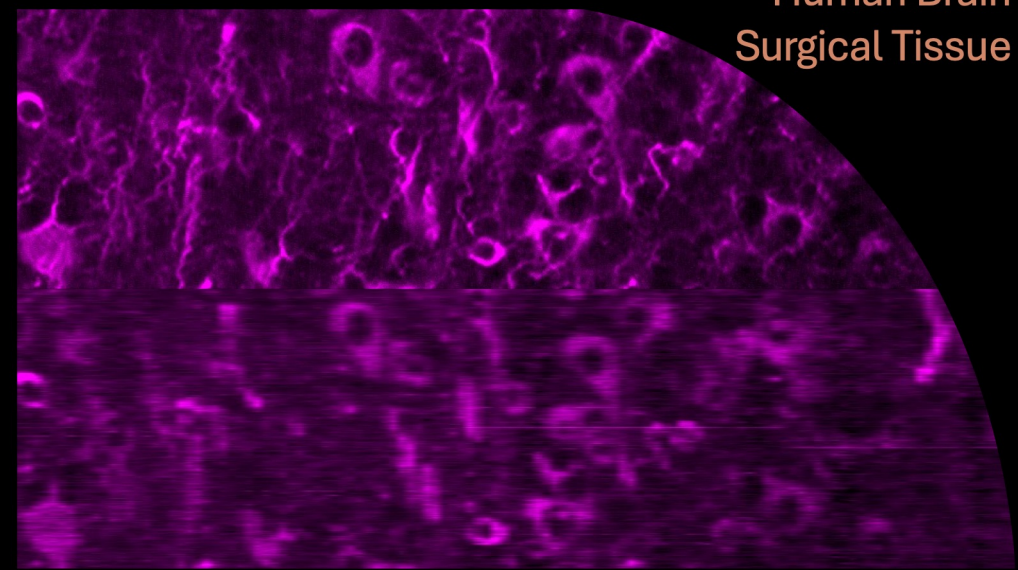
Translation to multi-clinical platform & modalities



Total Protein Stain
for Human Tissue



Human cell
Organoids



Human Brain
Surgical Tissue

Future work Toward 3D Nanoscale Pathology of Parkinson's Disease

Target disease: Parkinson's disease is characterized by progressive dopaminergic neuron loss and Lewy body accumulation — features that are inherently 3D and span multiple brain regions.

Proposed approach: Apply pan-ExM with isotropic enhancement to human brain tissue from Parkinson's patients, enabling whole-region 3D visualization of dopaminergic neuron morphology.

Long-term goal: Build a multi-scale atlas linking radiological biomarkers to nanoscale tissue changes, unleashing the diagnostic potential of clinical imaging protocols.

Thank you to my mentors and collaborators



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江安世
CMU

Pin-Hsun Lian, Tzu-Yi Chuang, Ya-Ding Liu, Li-An Chu, Sheng-Cheng Chang, Yu-Chen Kuo, Wei-Kun Chang, Ann-Shyn Chiang, Gary Han Chang (2025, Dec). Self-Supervised Isotropic Resolution Enhancement of Expansion Microscopy via Quantized Compression. Journal of Imaging Informatics in Medicine, Accepted.

Gary Han Chang, Meng-Yun Wu, Ling-Hui Yen, Da-Yu Huang, Ya-Hui Lin, Yu- Ru Luo, Ya-Ding Liu, Bin Xu, Kam W. Leong, Wen-Sung Lai, Ann-Shyn Chiang, Kuo-Chuan Wang, Chin-Hsien Lin, Shih-Luen Wang, Li-An Chu (2023, Dec). Isotropic Multi-Scale Neuronal Reconstruction from High-Ratio Expansion Microscopy with Contrastive Unsupervised Deep Generative Models. Computer Methods and Programs in Biomedicine